SMART ATTENDANCE SYSTEM USING FACE RECOGNITION

(Three-Tier Architecture Based Project)

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1. Introduction

Attendance management is an important activity in educational institutions and organizations. Traditional attendance systems are manual, time-consuming, and prone to errors such as proxy attendance.

This project introduces an automated attendance system using face recognition technology. It uses computer vision techniques to identify individuals and mark attendance automatically.

2. Problem Statement

The existing attendance systems face several issues:

- Manual effort required for marking attendance
- Chances of errors and manipulation
- Proxy attendance problems
- Lack of real-time tracking

Hence, there is a need for an automated, accurate, and secure system.

3. Objectives

- To automate attendance using face recognition

- To reduce manual work
- To improve accuracy and reliability
- To maintain digital attendance records

4. Scope of the Project

This system can be used in:

- Schools and colleges
- Offices and organizations
- Training institutes

It can be further extended to cloud-based systems and mobile applications.

5. Technologies Used

- Python
- Django Framework
- OpenCV
- face_recognition Library
- SQLite Database
- HTML, CSS

6. System Architecture (Three-Tier Architecture)

The system follows a three-tier architecture:

6.1 Presentation Layer

- User interface developed using HTML and CSS
- Allows user interaction such as login and attendance viewing

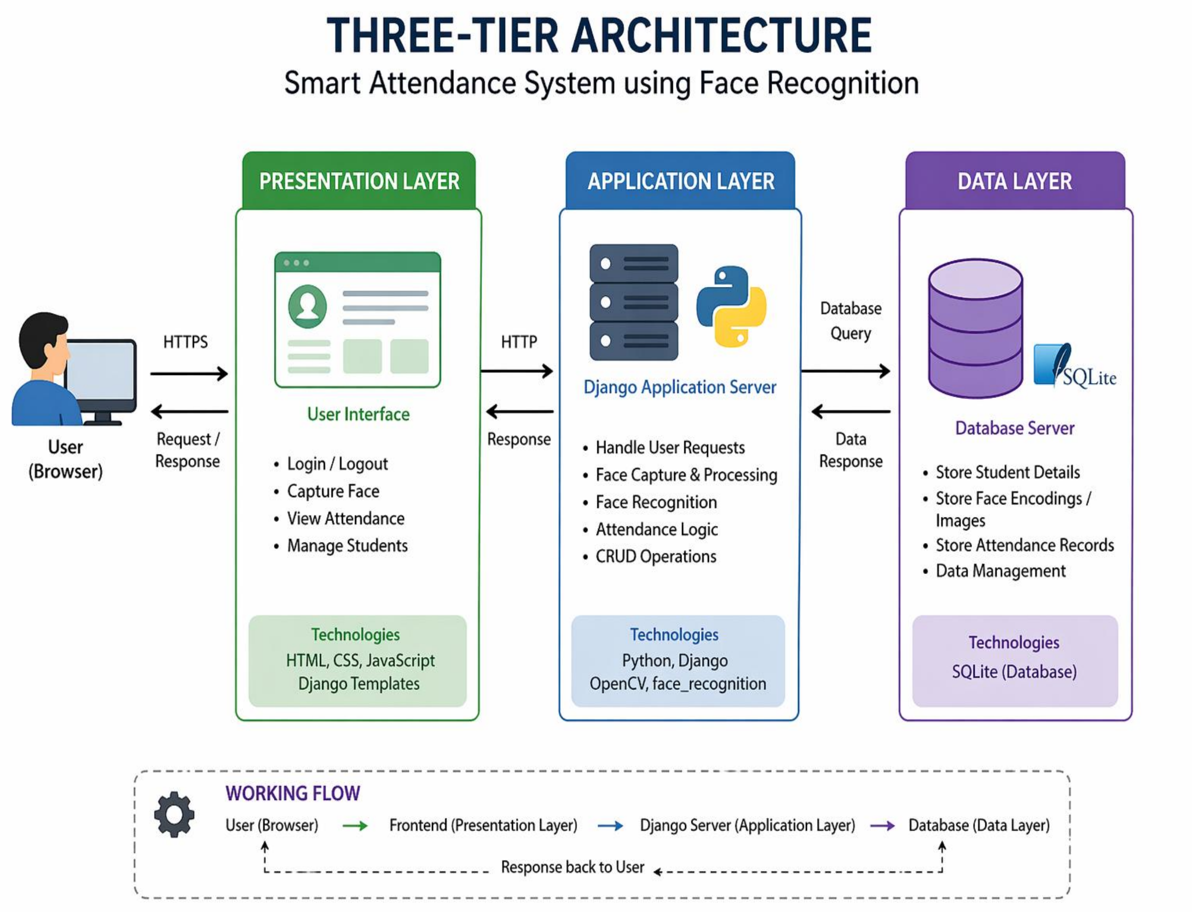
6.2 Application Layer

- Developed using Django (Python framework)
- Handles face recognition and attendance logic
- Processes user requests

6.3 Data Layer

- SQLite database
- Stores student details and attendance records

👉 Diagram:



7. System Design / Project Structure

The project is organized as follows:

- templates/ → Frontend
- app1/ → Application logic
- Project101/ → Django configuration
- db.sqlite3 → Database
- media/ → Stored images

8. Working of the System

1. User logs into the system
2. The system captures the face using webcam
3. Face encoding is generated
4. It is compared with stored data
5. If matched, attendance is marked
6. Data is stored in the database

9. Implementation Details

- Django is used as the backend framework
- OpenCV is used for image processing
- face_recognition library is used for face matching
- SQLite is used for data storage

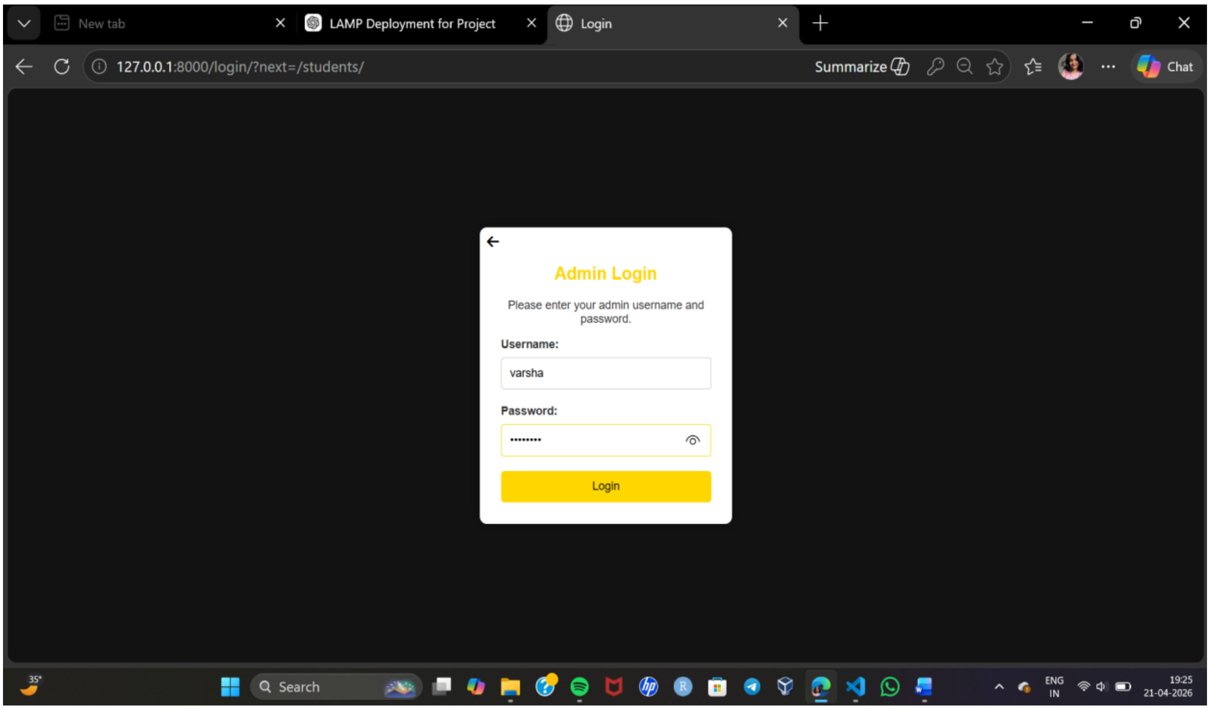
9.1 Steps To Run Project:

- python -m venv venv(first time only)
- venv\Scripts\activate
- pip install -r requirements.txt
- python manage.py makemigrations
- python manage.py migrate
- python manage.py createsuperuser(first time only)
- python manage.py runserver
- http://127.0.0.1:8000/ (open in broswer)
- For Admin Panel: http://127.0.0.1:8000/admin/

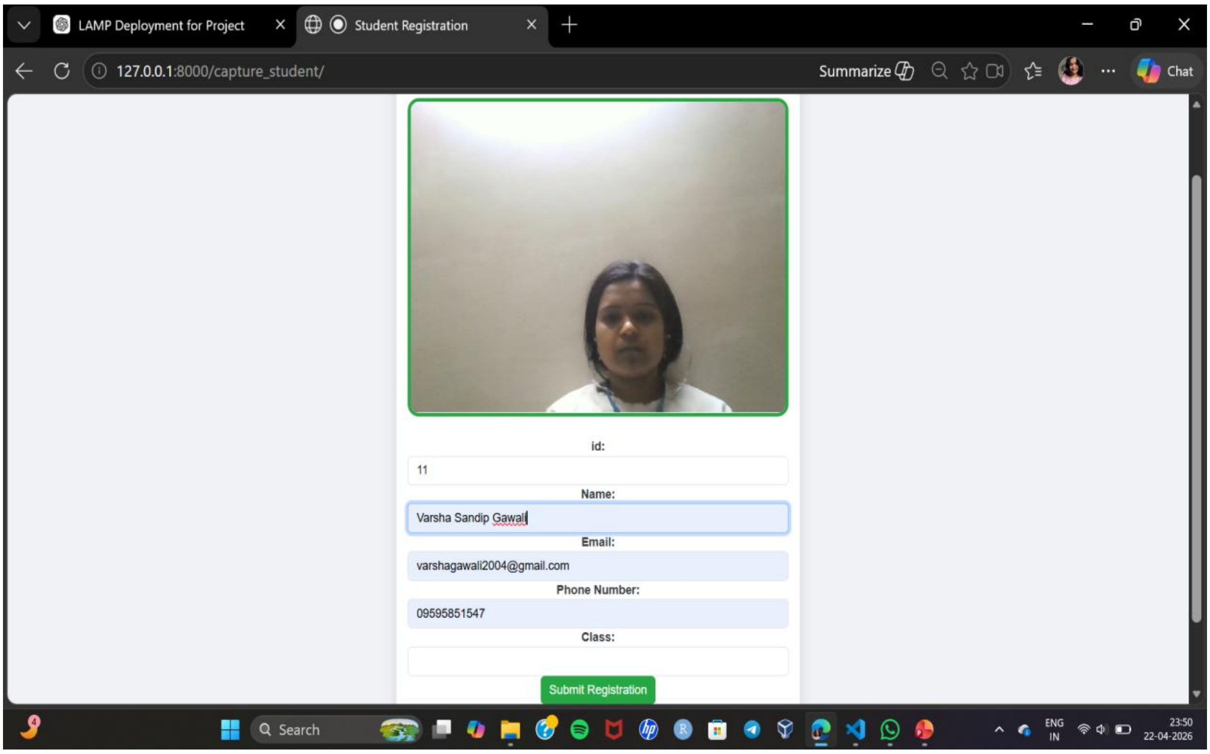
10. Results and Screenshots

👉

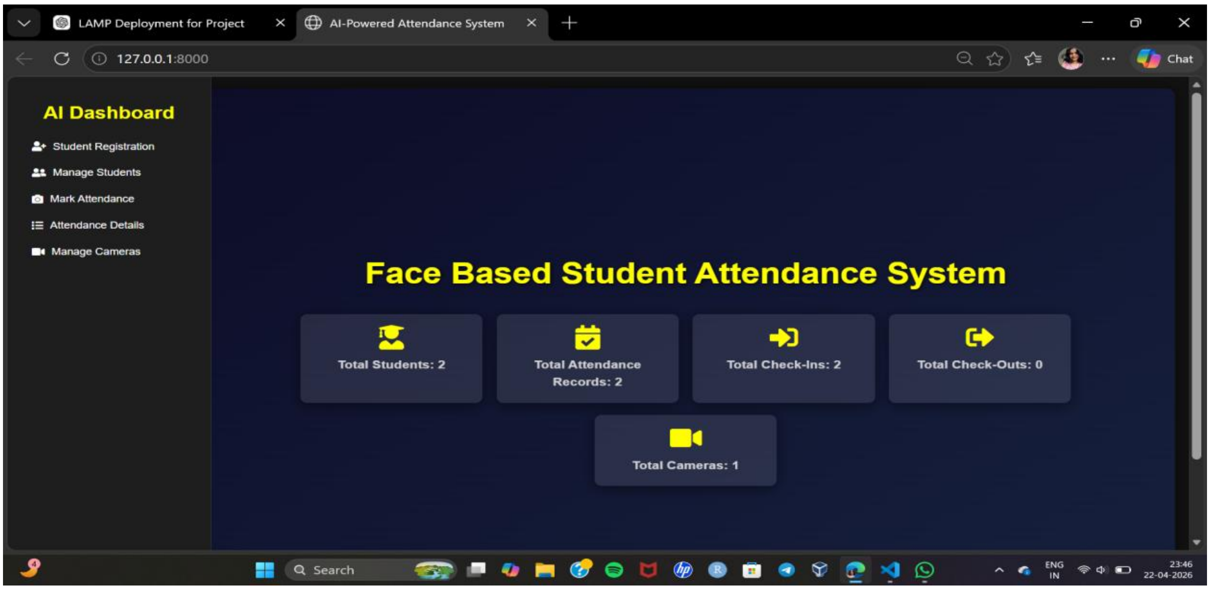
- **Admin Login page**



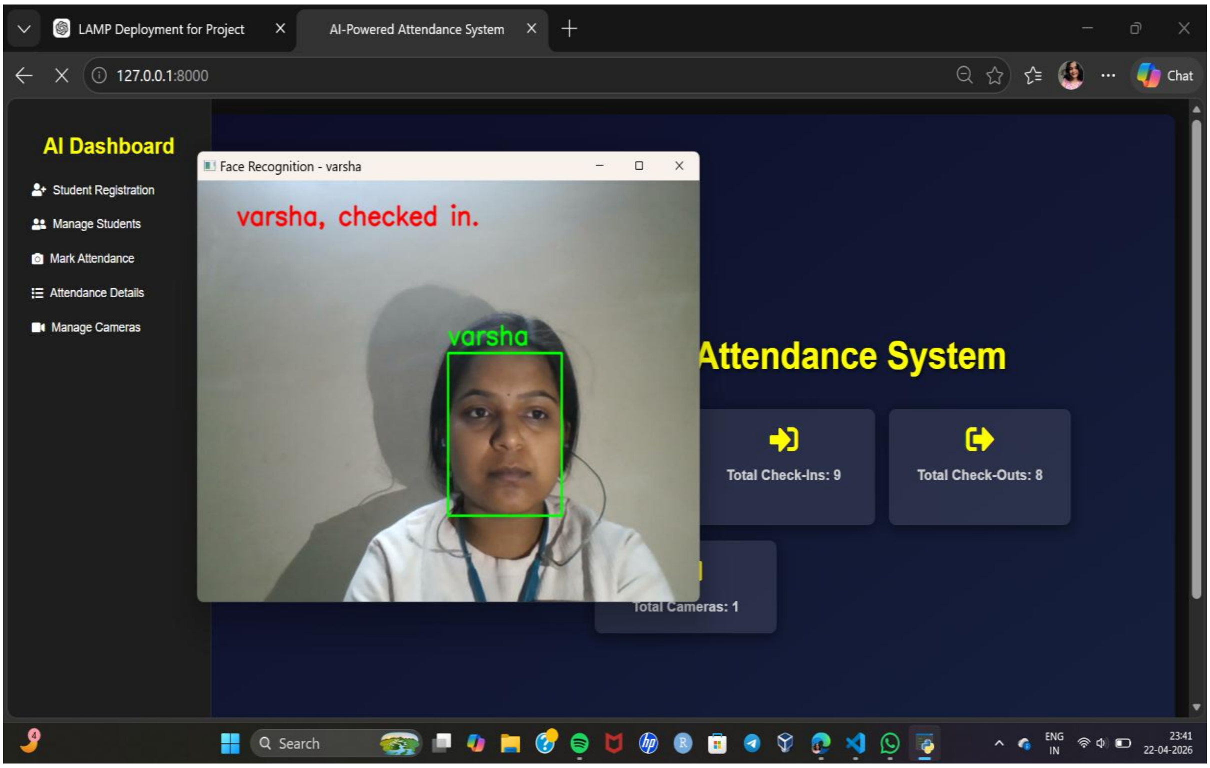
- **Student Registration Page**

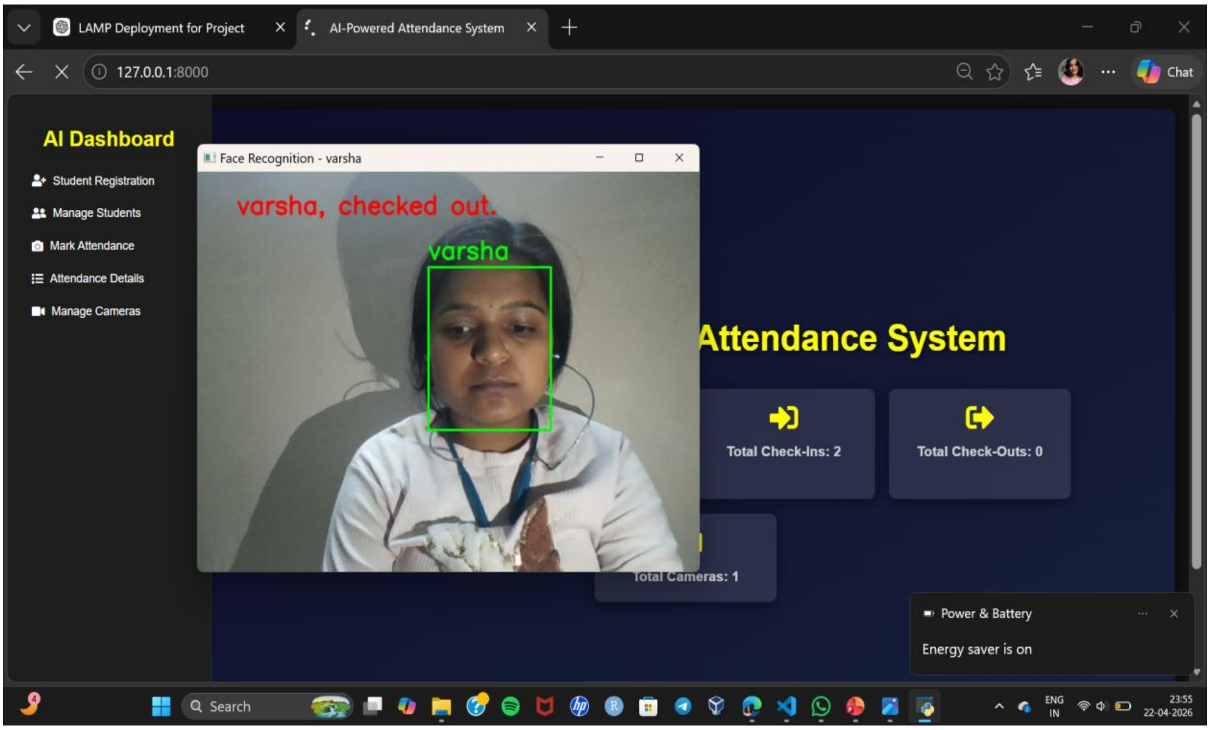


- **Home Page**

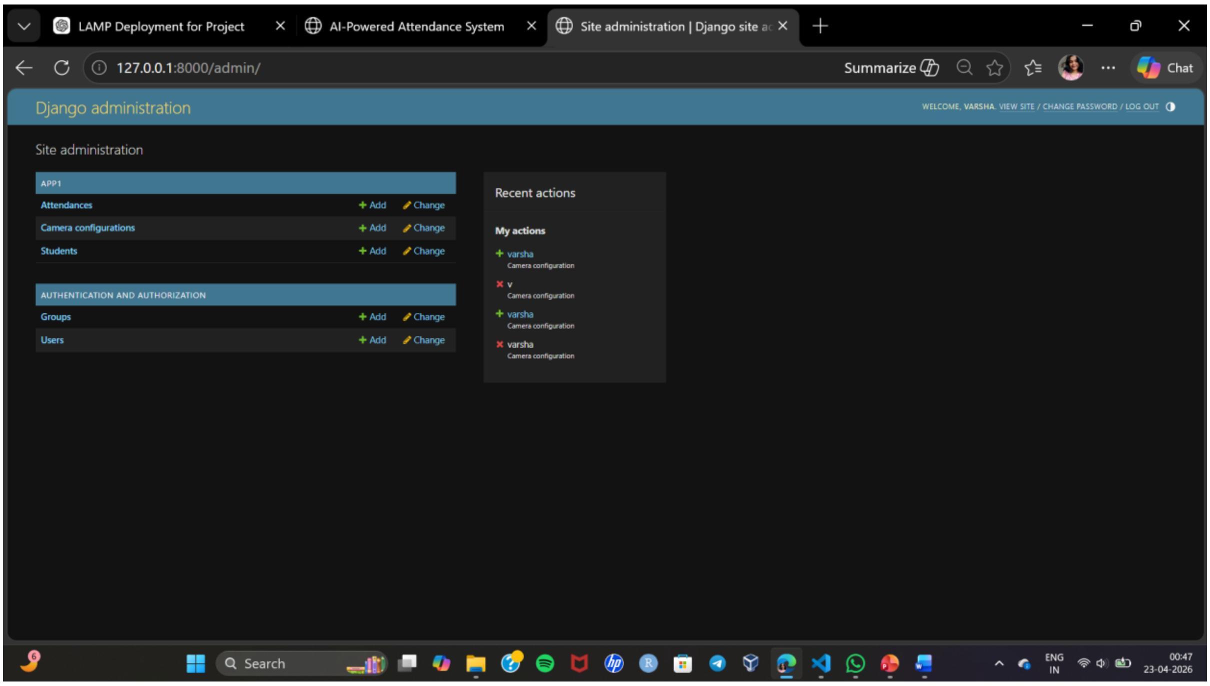


- **Attendance marking(checked-in and checked-out) Page**

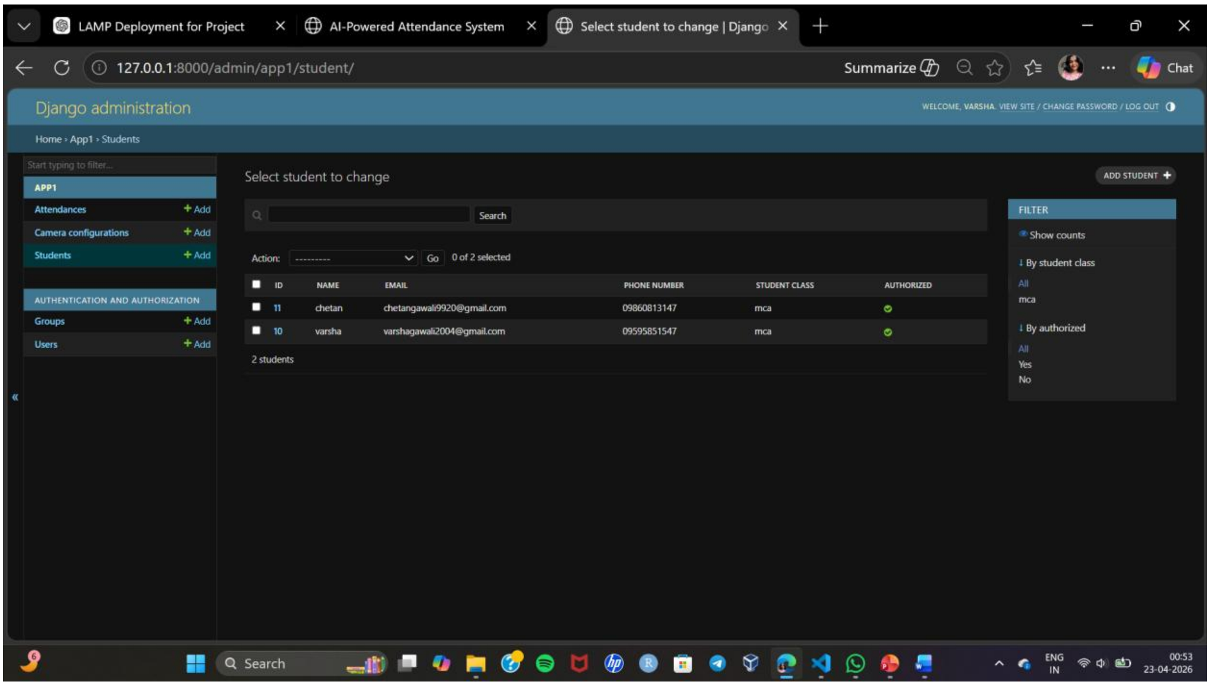




- **Admin panel**



- Database



11. Advantages

- Automated attendance system
- Reduces manual errors
- Prevents proxy attendance
- Easy to use and manage

12. Limitations

- Requires good lighting conditions
- Depends on camera quality
- Initial setup required

13. Future Scope

- Integration with cloud platforms like AWS
- Mobile application support
- Real-time analytics dashboard
- Multi-user and multi-location support

14. Conclusion

The project successfully demonstrates an automated attendance system using face recognition. It improves efficiency, accuracy, and security compared to traditional methods. The system follows a three-tier architecture and can be scaled for real-world applications.

15. References

- Django Documentation
- OpenCV Documentation
- face_recognition Library Docs
- Online tutorials and resources